

Multi-Class Hate Speech Detection in Sinhala-English Code-Mixed Social Media Text with Emoji Feature Integration Using XLM-RoBERTa

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Abstract—Hate speech detection in low-resource, code-mixed languages remains an open challenge, particularly for Sinhala-English social media text where prior work has been limited to binary classification, has discarded emoji information during preprocessing, and has been restricted to a single platform. We address these three gaps by fine-tuning XLM-RoBERTa-base for three-class classification (hate, offensive, neutral) on Sinhala-English code-mixed comments collected from Facebook and YouTube, comparing a baseline model against a proposed model that encodes emoji as textual sentiment cues. We describe our data collection, annotation, and evaluation methodology in detail, including a rigorous data-quality audit process, and report our findings with an explicit discussion of the failure modes and limitations encountered.

Index Terms—Hate speech detection, code-mixed text, Sinhala-English, XLM-RoBERTa, emoji features, low-resource NLP, social media.

I. INTRODUCTION

With the growing popularity of social media, hate speech detection in low-resource languages has emerged as an urgent research problem. In Sri Lanka, Facebook and YouTube provide large volumes of user-generated, code-mixed Sinhala text written in Roman characters interspersed with English. Muthuthanthri and Smith [1] built an initial baseline dataset of 2,205 labelled Sinhala-English Facebook comments and evaluated nineteen machine learning algorithms for hate versus non-hate classification, with BERT achieving 82% accuracy and a 90% ROC-AUC. Liyanage and Jayakumar [2] similarly observed a performance ceiling for classical machine learning models on Romanised Sinhala text, and Hettiarachchi et al. [3] confirmed that pre-trained embeddings outperform bag-of-words representations for this task.

Muthuthanthri and Smith [1] explicitly identified three limitations for future work: integrating emoji and symbol information into detection models, developing methods to counter symbol-based evasion, and extending data collection beyond Facebook to other platforms such as YouTube. To our knowledge, no subsequent study of Sinhala-English code-mixed hate speech has addressed these three gaps together. This paper makes the following contributions:

- We extend the classification scheme from binary (hate / not-hate) to a three-class taxonomy (hate, offensive,

neutral) following Davidson et al. [5], distinguishing targeted hate speech from general offensive language that is not directed at an identity group.

- We integrate emoji information as an explicit input feature via Unicode-to-text description encoding, rather than discarding it during preprocessing as in [1].
- We collect data across two platforms (Facebook and YouTube) rather than a single platform, following the cross-platform generalisation argument supported by Vasantharajan and Thayasivam [6].

Our research question is: *does fine-tuning XLM-RoBERTa with emoji encoding lead to a statistically significant improvement in macro-averaged F1-score over a fine-tuned XLM-RoBERTa baseline without emoji encoding, on three-class Sinhala-English code-mixed data from Facebook and YouTube, evaluated across five stratified folds?* This question is falsifiable: if emoji features introduce noise, cause overfitting on the minority class, or fail to produce a significant macro-F1 improvement, the hypothesis is disproved.

The remainder of this paper is organised as follows. Section II reviews related work in Sinhala and code-mixed hate speech detection and multilingual Transformer models. Section III describes our data collection, annotation, preprocessing, and model architecture in detail. Section IV reports our results across both experimental conditions. Section V discusses these results, including a dedicated failure analysis and ethical reflection. Section VI concludes with our contributions in light of the results and directions for future work.

II. RELATED WORK

Early classical machine learning approaches to Sinhala and Singlish hate speech detection established meaningful but limited baselines. Hettiarachchi et al. [3] and Liyanage and Jayakumar [2] both found that ensemble classifiers over hand-crafted features plateau around 84% accuracy on Romanised Sinhala text, with no transformer-based methods explored in either study. Fernando et al. [4] showed that FastText combined with RNNs achieves strong AUC-ROC on Sinhala social media text, motivating the move toward learned embeddings over classical features.

Muthuthanthri and Smith [1] remain the closest prior work to ours: a Facebook-only, binary-labelled dataset evaluated with BERT, explicitly excluding emoji from the input. Davidson et al. [5] demonstrated, in the (monolingual) hate speech

literature more broadly, that collapsing hate and offensive speech into a single category measurably reduces classifier accuracy, motivating our adoption of their three-class taxonomy. Sap et al. [14] showed that hate speech classifiers trained on majority-dialect data systematically misclassify minority-dialect text, which directly informs our ethical considerations regarding potential bias against Tamil and Muslim community speech in the Sri Lankan context.

On the modelling side, Conneau et al. [7] introduced XLM-RoBERTa, pretrained on 2.5TB of multilingual Common Crawl data across 100 languages, and Vasantharajan and Thayasivam [6] demonstrated its superiority over BERT and SVM baselines on the linguistically similar task of Tamil-English code-mixed YouTube comment classification – directly motivating both our choice of XLM-RoBERTa and our inclusion of YouTube as a data source. Mozafari et al. [11] further showed that models pretrained on web-crawled text outperform those trained on curated corpora for social media classification tasks, reinforcing the choice of XLM-RoBERTa (Common Crawl-pretrained) over standard BERT. Barbieri et al. [12] established macro-F1 as the standard metric for imbalanced multi-class tweet classification benchmarks, which we adopt as our primary metric.

Across this body of work, the most advanced prior Sinhala-English system – Muthuthanthri and Smith [1] – remains constrained to binary classification, discards emoji during preprocessing, and draws from a single platform. No subsequent study combines a three-class taxonomy, emoji-aware input encoding, and cross-platform data collection for this language pair; this is the specific, compound gap our work addresses.

III. METHODOLOGY

A. Data Collection

Data was collected from two social media platforms: Facebook, via the Facepager tool interfacing with the Facebook Graph API, and YouTube, via the YouTube Data API v3. Collection targeted public comment threads on political, religious, and news-related content between commentary. Only publicly available posts and comments were collected; private groups, individual pages, and direct messages were excluded.

B. Annotation

Each collected comment was independently labelled following a written annotation guideline defining three classes – hate speech, offensive language, and neutral content – with worked examples drawn from prior literature [1], [5].

C. Preprocessing

Comments were preprocessed via URL removal, username/mention removal, emoji extraction and encoding (or removal, for the baseline condition), lowercasing of English tokens, repeated-character normalisation, and whitespace normalisation, with a minimum length filter of three tokens applied after preprocessing. Emoji were converted to their Unicode text description (e.g. an angry-face emoji becomes the text “angry face”) using the Python `emoji` library and

appended inline to the comment text, allowing the model to learn emoji meaning jointly with surrounding context rather than treating it as noise.

D. Data Quality Audit

Prior to model training, the annotated corpus was audited for exact and near-duplicate content using core-text clustering (filler-word and emoji-stripped normalisation followed by exact and TF-IDF-based near-duplicate grouping). This step is critical for any code-mixed social-media hate speech corpus, particularly given the risk that near-duplicate or templated content can trivially inflate held-out performance metrics through train/test leakage rather than genuine model generalisation – a failure mode we discuss in detail in Section V-A.

E. Model Architecture

The system fine-tunes XLM-RoBERTa-base [7] for three-class sequence classification. The only architectural difference between the baseline and proposed conditions is the emoji-handling step in preprocessing (Section III-C): the baseline strips emoji entirely, matching the approach of [1], while the proposed model encodes emoji as appended text tokens. All other architectural and training choices are held constant across both conditions so that any observed difference in macro-F1 can be attributed specifically to emoji handling. Table I summarises the training configuration.

TABLE I
TRAINING CONFIGURATION

Parameter	Value
Base model	xlm-roberta-base
Max sequence length	128 tokens
Optimiser	AdamW
Learning rate	2e-5
Dropout	0.1
Class weighting	Inverse-frequency balanced
Early stopping	Validation-loss based, patience 2

Class-weighted cross-entropy loss was used to address label imbalance, with weights computed as the inverse frequency of each class in the training fold. AdamW was selected over standard SGD because it decouples weight decay from the gradient update, which has been shown to improve generalisation when fine-tuning large pretrained Transformer models [7]; a learning rate of 2e-5 follows the range established as effective for fine-tuning BERT-family models on downstream classification tasks. XLM-RoBERTa was chosen over a convolutional or recurrent architecture because the task requires transferring pretrained multilingual representations to a low-resource, code-mixed setting with limited labelled data – a setting where Transformer-based transfer learning has been shown to substantially outperform architectures trained from scratch on code-mixed South Asian language data [6], [11].

F. System Architecture

Figure 1 shows the end-to-end pipeline: raw platform data is collected, annotated by independent human raters, passed

through the preprocessing pipeline described above, tokenised, and fed to the fine-tuned XLM-RoBERTa classifier, which outputs a three-class softmax distribution over {neutral, offensive, hate}.

Facebook + YouTube → Raw Corpus →
Annotation + Preprocessing → Feature Extraction
→ XLM-RoBERTa-base → Output Classification
→ Evaluation

Fig. 1. End-to-end system architecture for multi-class Sinhala-English code-mixed hate speech detection with emoji feature integration.

G. Baseline Definition

The baseline is an XLM-RoBERTa-base model trained identically to the proposed model in every respect except that emoji are stripped from the input rather than encoded. This isolates emoji handling as the sole experimental variable, allowing any difference in macro-F1 between the two conditions to be attributed specifically to emoji information.

H. Evaluation Protocol

Macro-averaged F1-score is the primary evaluation metric, selected because it weights all three classes equally regardless of frequency – appropriate given the class imbalance in hate speech corpora, where accuracy is a misleading metric [1], and consistent with its use as the standard metric for imbalanced multi-class tweet classification [12]. Secondary metrics include per-class precision, recall, and F1, macro-averaged AUC-ROC, and confusion matrices for failure analysis. Evaluation uses stratified k -fold cross-validation ($k = 5$) rather than a single fixed train/validation/test split, because with a target corpus of approximately 2,500 comments a single split would leave only approximately 375 comments for testing – too few to yield a stable per-class F1 estimate given the three-way class imbalance. Five-fold cross-validation instead produces five independent test estimates, allowing performance to be reported as a mean and standard deviation that reflects genuine variability rather than the outcome of one arbitrary split, with stratification ensuring the class distribution is preserved in every fold. The identical fold assignment is reused for both the baseline and proposed conditions to enable a paired statistical comparison. Statistical significance between the two conditions' per-fold macro-F1 scores is assessed using the Wilcoxon signed-rank test, chosen over a paired t -test because it does not assume normally distributed differences, which is appropriate given the small number of paired observations.

IV. RESULTS

TODO: Report the Wilcoxon signed-rank test result here (W statistic, p -value, and whether $p < 0.05$), and state plainly if the result is not statistically significant – this is a valid, falsifiable outcome per the research question in Section I, not a result to hide or obscure.

TABLE II
MACRO-F1 BY CONDITION (TODO: REPLACE WITH REAL VALUES)

Condition	Mean Macro-F1	Std. Dev.
Baseline (no emoji)	TODO	TODO
Proposed (with emoji)	TODO	TODO

V. DISCUSSION

A. Failure Analysis

During development, a critical failure mode was identified and is worth reporting transparently as part of this paper's methodological contribution: an initial dataset later found to contain templated, near-duplicate content (Section III-D) produced a macro-F1 of 1.0000 across every cross-validation fold, for both the baseline and proposed conditions, with zero measurable difference between them. This was diagnosed, not merely observed: a trivial bag-of-words logistic regression model, entirely incapable of contextual language understanding, achieved the identical ceiling score on the same data, confirming that the near-perfect performance reflected the dataset's narrow, near-deterministic per-class vocabulary rather than genuine model competence. This finding is reported here as a methodological caution for future Sinhala-English code-mixed hate speech research: near-duplicate and template-generated content – whether from data augmentation or from repetitive real-world posting patterns – can silently invalidate cross-validation estimates unless explicitly audited for, and macro-F1 alone, without a duplication/leakage audit, is insufficient evidence of genuine classifier competence.

B. Ethical Reflection

This research handles sensitive content directly implicating protected and marginalised communities in Sri Lanka, including Tamil and Muslim minorities historically targeted during periods of communal unrest. Several ethical safeguards were built into the methodology from the outset. All data collection was restricted to publicly available posts and comments; private accounts, groups, and direct messages were excluded entirely. User identifiers – names, profile URLs, and usernames – were stripped prior to annotation and are not present in the final dataset, consistent with data minimisation principles. Annotators were informed of the potential psychological effects of prolonged exposure to hateful and offensive content and advised to limit annotation sessions to 30 minutes at a time.

A specific bias risk, documented by Sap et al. [14], is that hate speech classifiers trained predominantly on majority-dialect data can systematically misclassify minority-dialect or minority-community speech – for example, flagging in-group reclaimed language or minority-community political speech as hate speech at disproportionate rates. Clear annotation guidelines defining each class with worked examples were used specifically to reduce annotator subjectivity and mitigate bias against minority-community speech, though this risk cannot be eliminated through guidelines alone and would require targeted fairness auditing (e.g., per-community false-positive-rate analysis) as future work before any real-world deployment.

Finally, the Online Safety Act No. 9 of 2023 (Sri Lanka) creates a regulatory context in which automated content moderation tools for code-mixed Sinhala-English text may see real-world deployment; the authors caution against deploying any model resulting from this research in a production content-moderation context without further fairness auditing, human-in-the-loop review of flagged content, and an appeals mechanism for affected users, given the demonstrated risk of disproportionate impact on minority communities.

C. Limitations

- Facebook data collection was constrained by Meta's post-2018 Graph API access policy, which requires Page-admin access or App Review for comment-level data beyond a developer's own pages;
- The final annotated dataset size (smaller than the original target of 2,500 comments, reflecting the time constraints of a single-semester capstone project; this limits the statistical power of the five-fold cross-validation design, particularly for the Wilcoxon signed-rank test (with $k = 5$ paired folds, the smallest achievable two-sided p -value is 0.0625, meaning even a consistent effect across all folds may not reach conventional significance);
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VI. CONCLUSION

This paper presented a methodology for multi-class Sinhala-English code-mixed hate speech detection with emoji feature integration, addressing three gaps identified in prior work: binary-only classification, emoji removal during preprocessing, and single-platform data collection.

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